

Project No.	Title	Description / Task	Expected Output
1	Image Transformation Gallery	Apply 2×2 matrices (scaling, rotation, shearing, reflection) to digital images. Explain determinant & eigenvalues.	A set of original & transformed images with analysis.
2	Secret Code with Linear Transformations	Use invertible matrices as keys to encode & decode secret messages. Show why singular matrices fail.	Encoded/decoded messages with math explanation.
3	Linear Transformation in Computer Graphics	Simulate 2D game object motions using transformations (translation, rotation, reflection).	Simple animation or simulation demo.
4	Music and Sound Transformation	Represent audio as vectors; apply scaling, reflection, reversal, filtering. Distinguish which are linear.	Transformed sound clips with explanation.
5	Symmetry and Reflection in Art	Use reflections and rotations to design geometric art/tessellations.	Creative artwork + math justification.
6	Exploring 3D Transformations	Use software (GeoGebra, Python, Blender) to apply rotation, scaling, projection to 3D objects.	3D visualizations and analysis.
7	Eigenfaces & Facial Recognition	Research eigenfaces in AI. Apply transformations to sample images; discuss eigenvalues/eigenvectors.	Short demo & conceptual report.
8	Transformation of Maps	Apply scaling, rotation, shear to maps. Show distortion effects.	Original vs. transformed map images with discussion.
9	Linear Transformation & Robotics	Model robot arm joint movements as transformations; show how composition gives final position.	Diagram/simulation of robot arm motions.
10	Designing a Linear Transformation Game	Build a quiz/game where players guess the transformation applied to shapes.	Working prototype (digital or paper-based).